SunD

Appendix: evidence, cost model, KPIs and visuals

Every figure carries a linked source. Figures marked ASSUMPTION are the team's own and are listed in A18.

Team UrbanByte · Chia Cai Hong (leader) · Tae Jie Hui · Chan Kin Chung

WHY IT IS NEEDED

A1 Demand and context evidence
A2 Demand and strategic fit

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A3 Visual concepts
A4 Prototype screens
A5 SunD Bay on site
A6 SunD Bay sizes

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A8 Regulatory fit
A9 SunD Balcony precedent

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MONEY, RULES AND FINE PRINT

A20 Who pays for Wallet credits
A21 Where a bay can legally go
A22 Revenue levers
A23 Main deck fine print

EVs outrun chargers 18 to 1, and over 1 in 5 homes has no roof.

EVs & CHARGING

122,286
battery EVs registered

Public charging bays	**6,904**
BEVs per bay	**≈ 18**
Charger target, 2030	**30,000**
Best-seller	**e.MAS 5**

HEAT

~1 million
homes pushed past 600 kWh

KL peak surface temp, 30 yrs	**+1.64 °C**
Subang, per decade	**+0.37 °C**
Level 1 heat-alert areas, Aug 2026	**17**
Heat-illness cases, Jan–Jul 2024	**112**

WHO IS LEFT OUT

22.8%
of homes are high-rise

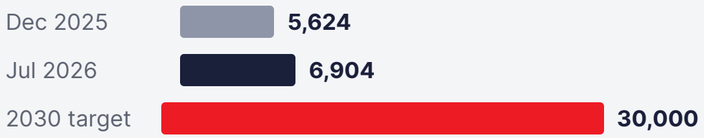
Selangor high-rise	**≈ 37%**
Aged 65+, Selangor	**8.7%**
Elderly admissions per +1 °C	**+12.9%**
Adults, no leisure activity	**~84%**

SIME DARBY

36,600
homes planned in Elmina

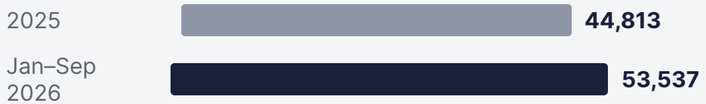
Area	**5,000 acres**
Elmina Ridge 1	**228 homes · 1 MWp**
Scope 1 & 2 by 2030	**−40%**
Net zero	**2050**

PUBLIC CHARGERS VS 2030 TARGET



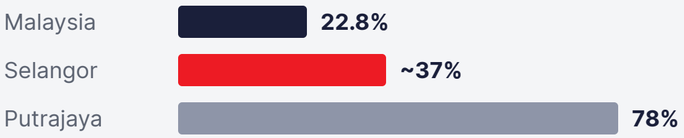
Gap to 2030: **23,000 chargers** in four years.

BEV REGISTRATIONS (JPJ)



9 months of 2026 already beat all of 2025.

HOMES THAT ARE HIGH-RISE



Every rooftop scheme starts from a roof the household controls.

PENINSULAR GRID 2025



0.74 kg CO_2 per kWh (2024), ~0.47 by 2030.

Sources: Kenanga Research, EV Day 2026, 122,286 BEVs (Oct 2026) · SoyaCincau/JPJ, EV registrations Sep 2026 · NST, 5,624 chargers at end-2025 (Feb 2026) · Lowyat.NET, 6,416 public chargers, 30,000 target (Jul 2026) · Business Today, 6,904 public charging bays (Sep 2026) · SoyaCincau, e.MAS 5 Malaysia's #1 EV (Aug 2026) · Bernama TV, ~1 million households past 600 kWh (Sep 2026) · Think City via Malay Mail, KL +1.64 °C over 30 years (Mar 2021) · Focus Malaysia/MetMalaysia, Subang +0.37 °C per decade (May 2026) · Miniandi et al., Earth Syst. Environ. 9, KL LST +2.2 °C 2013–2023 (2025) · Wang et al., Geoscience Letters 6:4, KL UHI ~8 °C modelled (2019) · The Star/ANN, 23 areas under Level 1 heatwave alert (Mar 2024) · Bernama, 17 areas under Level 1 heat status (Aug 2026) · Malay Mail/NADMA, 112 heat-illness cases, 5 deaths (Jul 2024) · The Star/MOH, 56 heat-related cases, 2 deaths (May 2026) · DOSM, Special Report: Living Quarters, Census 2020 (Apr 2026) · DOSM, Current Population Estimates 2026 (Jul 2026) · Yong et al., IJERPH 20(10):5910, heatwaves and Selangor elderly hospitalisations (2023) · MOH, NHMS 2023 Fact Sheet · CodeBlue, NHMS 2023, 84% no sport/leisure activity (May 2024) · SDP, City of Elmina · SDP, Sustainability Report 2024 (May 2025) · SDP, Integrated Annual Report 2025 (Apr 2026) · SDP, Net zero by 2050 pledge (Nov 2023) · Reuters/GSO via The Edge, 2025 generation mix (Jan 2026) · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026) · Single Buyer, GEF projections 2025–2030.

Green homes sell, chargers are scarce, and SDP has committed.

HOMEBUYERS
PropertyGuru Consumer Sentiment, 1H 2026

93%

of Malaysians willing or open to paying more for a home with green features



KEY POINT

"Free solar credits with your home" is a **pricing lever**, not a cost.

EV DRIVERS
EY Mobility Consumer Index Malaysia, Oct 2024

61%

of EV drivers cite finding a charger as a concern



KEY POINT

SunD Bay puts charging **where residents already park**.

SIME DARBY PROPERTY
its own published commitments

- **Dec 2023**
  MoU with TNB: rooftop solar, **EV charging** and microgrids across SDP townships
- **Apr 2025**
  First in Malaysia to join CREAM, starting in City of Elmina
- **Sep 2025**
  Residential solar under NETR **pilot in Elmina**

"a practical and scalable model to unlock solar generation capacity from within our townships"
SDP press release, 28 Apr 2025

KEY POINT

SunD is the **resident-facing layer** on SDP's own roadmap.

NATIONAL DIRECTION

80% xEV share of new cars by 2050 (NETR)

30,000 public chargers by 2030

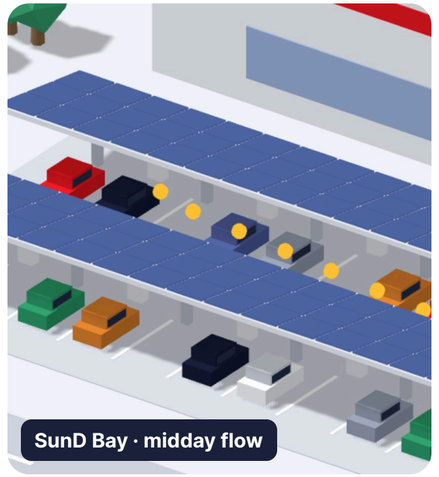
70% renewable capacity by 2050

Sources: Business Today, PropertyGuru 1H26 Consumer Sentiment Study (Sep 2026) · Carz, EY Mobility Consumer Index Malaysia (Oct 2024) · TechNode Global, TNB and Sime Darby Property MoU (Dec 2023) · SDP, Community solar rollout with TNB (Apr 2025) · SDP, Residential solar launch under NETR (Sep 2025) · SDP, Sustainability Report 2024 (May 2025) · Autobuzz, NETR targets 80% xEV share of TIV by 2050 (Aug 2023) · Lowyat.NET, 6,416 public chargers, 30,000 target (Jul 2026) · Argus Media, NETR: 70% RE capacity by 2050 (Aug 2023)

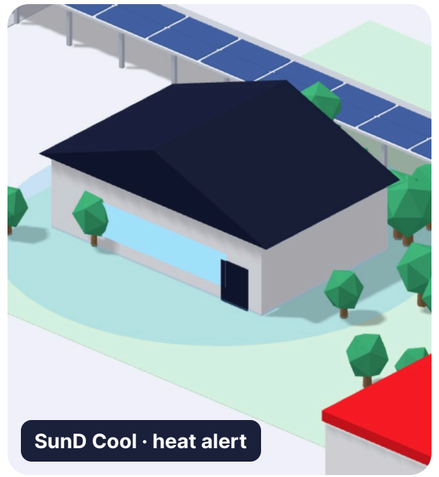
The SunD township in 3D.

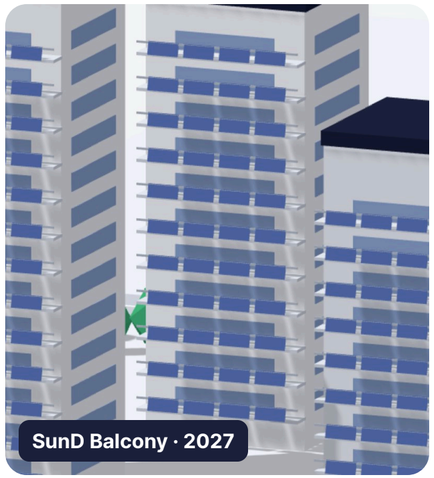


1 CREAM homes · roof hosts earn
2 SunD Bay · mall car park
3 Condos · slices + balcony kits
4 SunD Walk · shaded route to bus stop
5 SunD Cool · community hall
6 Bus stop / transit hub



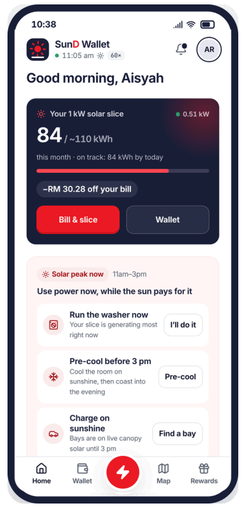




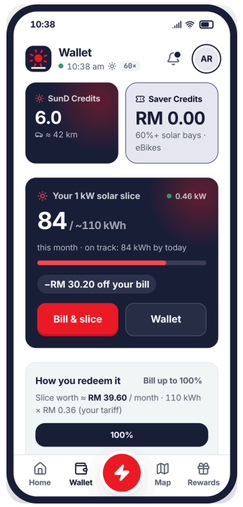


Sources: SDP, CREAM pilot in City of Elmina: roof hosts lease, power to SDP Innovation Park (Sep 2025) · Xu, Bardhan et al., Sci. Total Environ. 958 (2025)

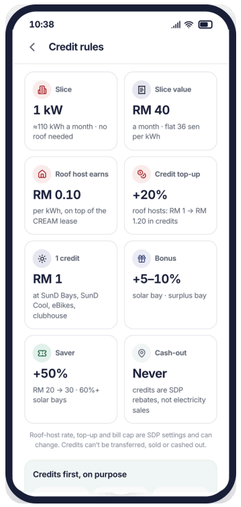
The working SunD Wallet prototype.



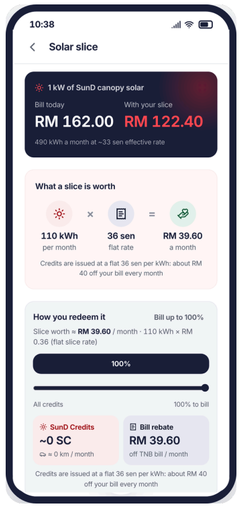
Home
slice, bill, nudges



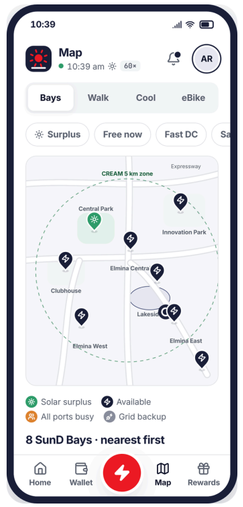
Wallet
credits + Saver



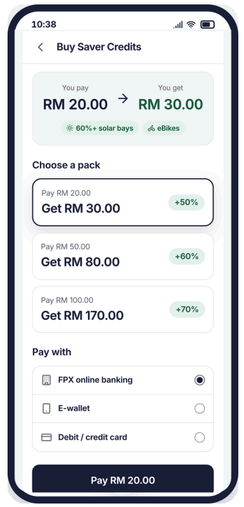
Credit rules
what SDP can tune



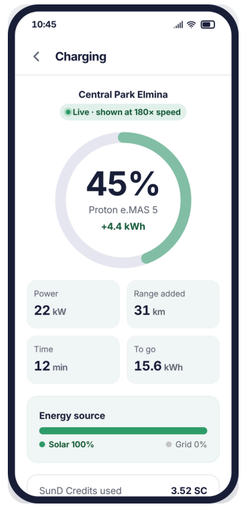
Solar slice
RM 162 → 122.40



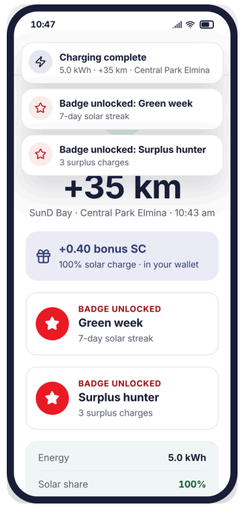
Green-bay map
solar stock live



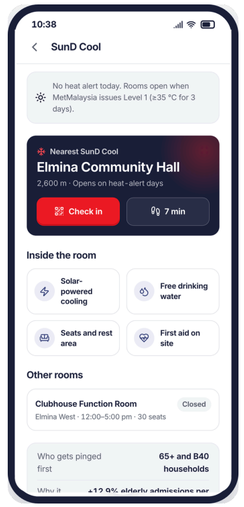
Saver Credits
RM 20 → RM 30



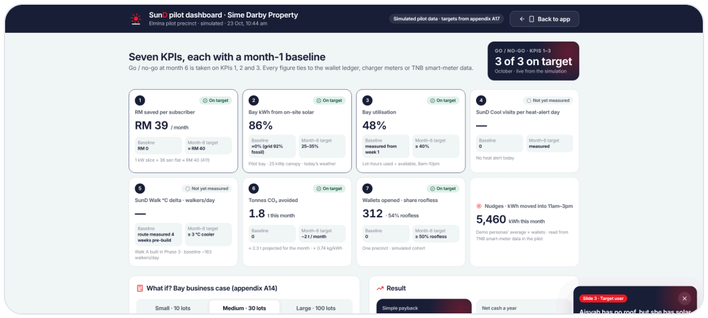
Charging
solar share live



Receipt
+35 km, 100% solar



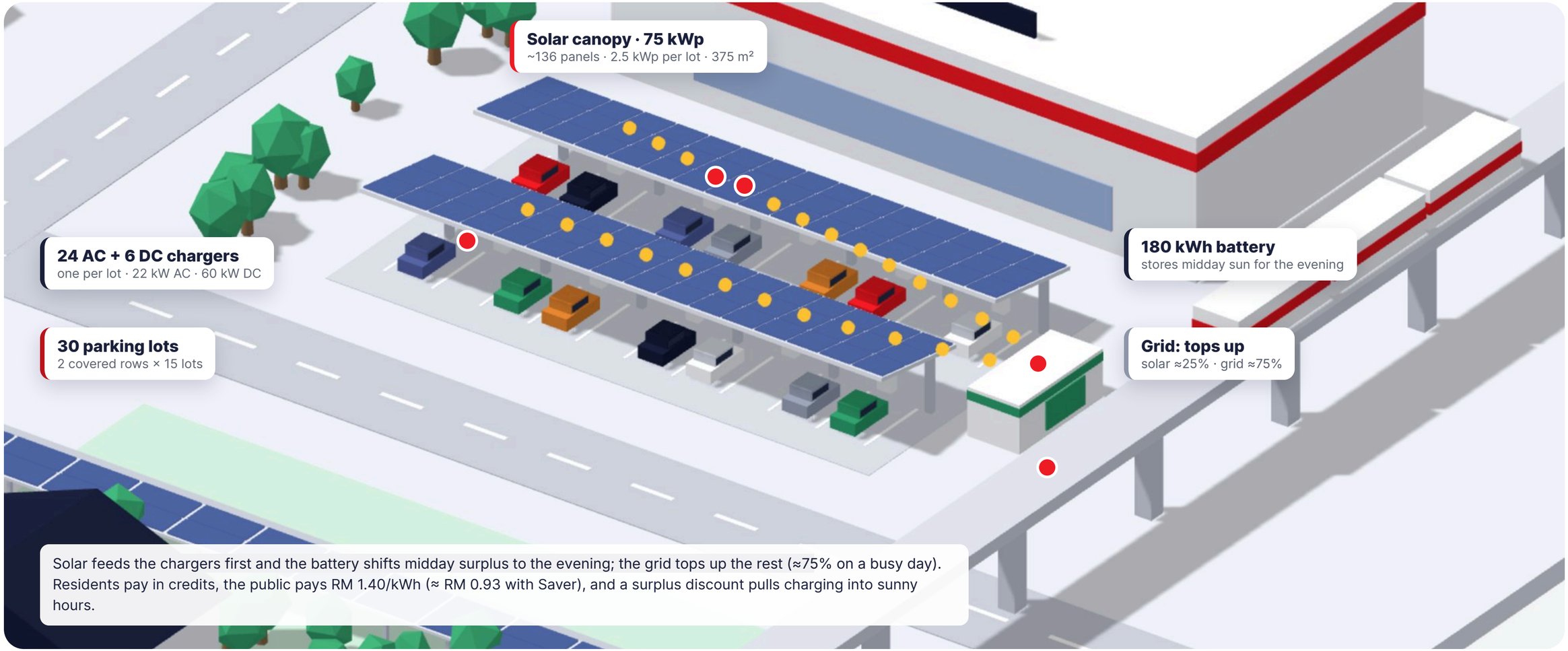
SunD Cool
nearest cool room



Plus the SDP pilot dashboard: 7 KPIs with baselines and a live what-if payback calculator.
Try it: chchia28.github.io/SD-Hackathon

Sources: SDP, City of Elmina

Inside a medium SunD Bay.



Solar feeds the chargers first and the battery shifts midday surplus to the evening; the grid tops up the rest (≈75% on a busy day). Residents pay in credits, the public pays RM 1.40/kWh (≈ RM 0.93 with Saver), and a surplus discount pulls charging into sunny hours.

Sources: Trexon, Solar carport: ~2.5 kWp per bay (2026) · Trexon, EV charger installation cost 2026 (Mar 2026) · EV Sifu, EV charging cost Malaysia 2026

Three bay sizes: clubhouse, mall and highway.

Small

10 lots

25 kWp · 8 AC + 2 DC · 60 kWh

RM 0.40–0.58M

before tax allowance · **RM 0.34–0.49M** after GITA

- 2–3 months to build (est.)
- clubhouse, park, condo car park
- ~90 kWh a day · ~24 t CO_2 a year

Medium

30 lots

75 kWp · 24 AC + 6 DC · 180 kWh

RM 1.19–1.73M

before tax allowance · **RM 1.02–1.47M** after GITA

- 3–4 months to build (est.)
- mall, commercial hub
- ~270 kWh a day · ~73 t CO_2 a year

Large

100 lots

250 kWp · 80 AC + 20 DC · 600 kWh

RM 4.0–5.8M

before tax allowance · **RM 3.4–4.9M** after GITA

- 4–6 months to build (est.)
- highway stop, flagship
- ~900 kWh a day · ~243 t CO_2 a year

UNIT COSTS USED

Canopy solar **RM 3,500–5,000 / kWp** · AC 22 kW **RM 8–15k** · DC 60 kW **RM 60–80k** · Battery **RM 1,500–2,000 / kWh** · Civil **+10%**

KEY POINT The small bay is the pilot's measuring tool; the **medium mall bay pays back in 4.7–6.8 years** at 2 sessions/lot/day, before Wallet credits (A14, A20).

Sources: Trexon, Solar carport: ~2.5 kWp per bay (2026) · Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Trexon, EV charger installation cost 2026 (Mar 2026) · MGTC/MyHIJAU, GITA asset list (Apr 2025) · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026)

Every part of SunD already works somewhere else.

AUSTRALIA
Allume SolShare

→ SunD Wallet slices

10,000
apartments in 4 countries share their building's solar (Apr 2026)

Bill cut per apartment
−34% · ~A$381 a year

Running in
Australia · Germany · UK · USA

KEY POINT
Households **without a roof** can share a building's solar.

UNITED KINGDOM
Octopus Saving Sessions

→ Wallet nudges & credits

~700k
households paid to shift their power use

Demand shifted
1.86 GWh in 13 sessions

Rewards paid
£5.3 million

KEY POINT
Paying **credits for shifting use** works at national scale.

FRANCE
Decree 2024-1023

→ SunD Bay

≥ 50%
of each car park over 1,500 m²: solar or green canopies

Applies to
Car parks over 1,500 m²

Deadline
Jul 2026 – Jul 2028

KEY POINT
Solar canopies over parking are becoming **law**, not novelty.

SINGAPORE
LTA Walk2Ride

→ SunD Walk

200 km
of sheltered walkways built

Coverage
Within 400 m of MRT stations

Budget
S$300 million

KEY POINT
A tropical city **shades the walk** to transit as policy.

INDIA · AHMEDABAD
Heat Action Plan

→ SunD Cool

1,100+
deaths avoided a year

Hottest days (≥ 45 °C)
Mortality −27%

Running since
2013

KEY POINT
A city **heat-response plan** measurably saves lives.

✓ Every SunD part copies a model that already works. The new part is **joining all five in one township wallet**, run by the developer that owns the roofs, car parks and halls.

Sources: pv magazine Australia, Allume SolShare reaches 10,000 apartments (Apr 2026) · Solar Power Portal, Octopus Saving Sessions displace almost 2 GWh (Apr 2023) · pv magazine, France makes solar mandatory on parking areas, Decree 2024-1023 (Nov 2024) · LTA Singapore, 200 km of sheltered walkways under Walk2Ride (Sep 2018) · NRDC, Ahmedabad Heat Action Plan evidence (Jun 2019; J. Environ. Public Health 2018)

Every scheme SunD uses is in force today, except balcony solar.

CREAM

IN FORCE

15 → 9 sen

access charge per kWh (Aug 2025)

- Sells roof output within **5 km**
- Elmina pilot: **SDP is the aggregator**

SUND USES IT

Hosts near a bay earn **SunD credits**; their power still goes to Innovation Park.

Solar ATAP

IN FORCE

≤ 1 MW

non-domestic cap, from 1 Jan 2026

- s.7.2: **car parks, walkways** on the same site
- Multi-tenant premises excluded

SUND USES IT

Single-owner **SDP sites**; mall bays run zero-export (**A21**).

CRESS

BENCHMARK

14–18 sen

firm grid access charge per kWh

- Firm: 18 sen to end-2027, then 14 sen
- Non-firm **40 sen**

SUND USES IT

Not used: it is the grid cost SunD avoids.

Credits

NO LICENCE

0 kWh

sold or traded by SunD

- **SDP-funded rebates**, not power sales
- TNB stays every home's supplier

SUND USES IT

Loyalty-style; Saver packs: **BNM check**.

GITA

IN FORCE

100% → 60%

tax allowance; the 2027 cut is a Budget proposal

- Solar for own use: **60%**
- Offsets **70%** of statutory income

SUND USES IT

About **RM170–256k** tax saved (A13).

GET / mREC

OPTIONAL

3–5 sen

green tariff premium per kWh

- mRECs + AFA exemption
- Solar I-REC ≈ **US$3.45/MWh**

SUND USES IT

Optional: RECs for SDP's **Scope 2**.

Balcony solar

2027

No rules

Malaysian framework today

- SELCO needs competent-person installs
- **EC sandbox planned for 2027**

SUND USES IT

SDP pilot inside the sandbox (A9).

THE READING

All 3 things the pilot needs are in force today.

- ✓ Roofs → **CREAM**
- ✓ Canopies → **ATAP or zero-export**
- ✓ Credits → **rebates**
- Balcony → **2027**

Sources: Energy Commission, SAC/CAC reduced by up to 40% (Aug 2025) · The Edge, Govt sets guidelines for rooftop solar leasing (Mar 2025) · SDP, Residential solar launch under NETR (Sep 2025) · The Edge, Empowering homes with renewable energy (CREAM 2 MWp per feeder) · SDP, Sustainability Report 2024 (May 2025) · Energy Commission, Guidelines for Solar ATAP (in force 1 Jan 2026) · pv magazine, Malaysia launches Solar ATAP (Jan 2026) · Bernama, CRESS Acceleration Package, SAC 14 sen/kWh (Sep 2026) · The Edge, CRESS SAC now 18 sen/kWh (Oct 2026) · Business Today, non-firm SAC remains 40 sen (Sep 2026) · SEDA, Malaysia's 1st pilot run of P2P energy trading (2020) · MGTC/MyHIJAU, GITA asset list (Apr 2025) · SoyaCincau, Budget 2027 cuts EV-charging GITA to 60% (Oct 2026) · The Edge, GET premium cut to 3–5 sen/kWh (Jul 2025) · S&P Global, Malaysian solar I-REC US$3.45/MWh (Apr 2026) · Energy Commission, SELCO Guidelines (2025) · The Star/Bernama, EC eyes 2027 regulatory sandbox (Jun 2026) · CAP via Aliran, Balcony solar panels for clean energy transition (27 Aug 2026). Scheme numbers verified 23 Oct 2026.

Balcony solar: 1.5 million in Germany, none yet in Malaysia.

THE GERMAN PRECEDENT

1.53 M
plug-in systems running in Germany

€300–500
for an 800 W kit, bought by the household

800 W
plugs into a normal socket

~560 kWh
a year per kit (German yield)

ONE KIT, ONE MONTH, IN MALAYSIA

~47 kWh
a month, to measure

×

35–52 sen
own tariff

=

RM 16–24
off the bill

PROPOSED SANDBOX DESIGN · SDP PILOT 2027

NO FRAMEWORK TODAY · CAP ASKED · AUG 2026 · EC SANDBOX · 2027

Scope	One Elmina condo tower · **50–100 units**
Safety	Certified kits only · **zero export**
Strata	Model by-law via the JMB/MC; renters need landlord OK
Wallet link	**Bonus credit** per kWh generated
Evidence	Yield, safety, bill saving, uptake

22.8% of homes are high-rise: **no rooftop scheme** reaches them.

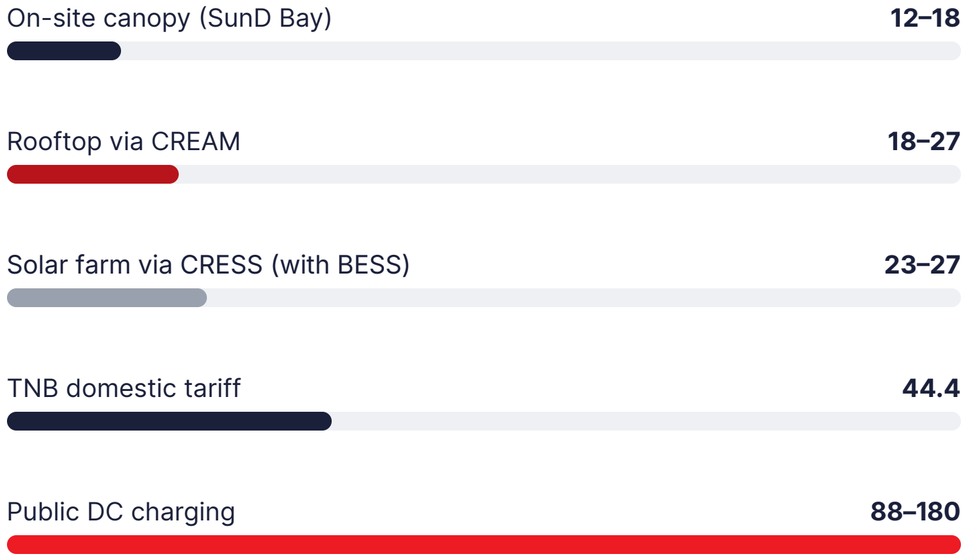
Sources: Bundesnetzagentur MaStR via kleineskraftwerk.de, 1.53 M systems (Oct 2026) · top agrar, 1 million plug-in systems (Jun 2025) · MonsterDealz, Balkonkraftwerk 2026: costs and yield (citing Verbraucherzentrale) · Energy Commission, SELCO Guidelines (2025) · CAP via Aliran, Balcony solar panels for clean energy transition (27 Aug 2026) · The Star/Bernama, EC eyes 2027 regulatory sandbox (Jun 2026) · DOSM, Special Report: Living Quarters, Census 2020 (Apr 2026). Sandbox design is the team's proposal, not an existing regulation.

From solar panels to kilometres driven.

WHERE	SIZE	kWh / DAY	km / DAY	50 km TOP-UPS
1 panel	550 W	2	14	¼
1 bay lot	2.5 kWp	9	63	1
1 Elmina roof	**4.4 kWp**	**16**	**110**	**2**
Small bay · 10 lots	25 kWp	90	630	13
Medium bay · 30 lots	75 kWp	270	1,890	38
Large bay · 100 lots	250 kWp	900	6,300	126
1 MWp (Ridge 1 size)	**1 MWp**	**3,600**	**25,200**	**504**

3.6 kWh per kWp a day (KL) · e.MAS 5 ≈ 7 km per kWh real-world · a 1 MWp array (Elmina Ridge 1's size) ≈ 25,000 km a day.
Why it matters: one Elmina roof covers about two 50 km top-ups a day; a 1 MWp array could cover ~500.

COST TO DELIVER 1 KWH · SEN



On-site canopy skips access charges and grid losses; CRESS without BESS ≈ 47–49 sen.

KEY POINT One Elmina roof covers **about two 50 km top-ups a day**; a 1 MWp array could cover ~500.

Sources: PVGIS SARAH3 / NASA POWER via Sunveyor: KL 1,402 kWh/kWp/yr · Carz, Proton e.MAS 5 launch (Oct 2025) · SDP, Sustainability Report 2024, p.42 (May 2025) · Energy Commission, SAC/CAC reduced by up to 40% (Aug 2025) · The Edge, CRESS SAC now 18 sen/kWh (Oct 2026) · SoyaCincau, TNB domestic tariff structure July 2025 · EV Sifu, EV charging cost Malaysia 2026

A 1 kW slice is worth about RM40 a month to a home with no roof.

WHAT A 1 kW SLICE IS WORTH

1 kW slice × **110 kWh** a month × **36 sen** flat credit rate = **≈ RM 40** a month

WHAT ONE BAY LOT DELIVERS

2.5 kWp per lot × **3.6 kWh** per kWp a day = **~9 kWh** a day → **~60 km** of e.MAS 5 driving

TNB RATE VS THE 36 SEN CREDIT



+ 4.55 capacity = **44.43 sen base**

≤ 600 kWh

30–36 sen

after EEI rebate, by band

> 600 kWh

~52 sen

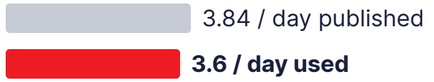
+ AFA + 8% SST

Threshold is temporarily **800 kWh** until 31 Dec 2026.

KL SOLAR YIELD

1,402 kWh

per kWp a year

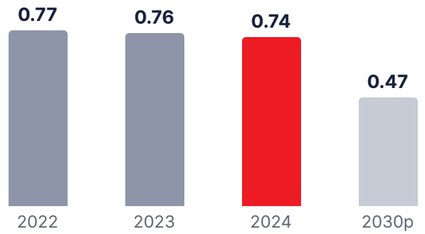


≈ **110 kWh** per kW a month, conservative.

GRID CO_2 PER kWh

0.74 kg

2024 actual, used in the deck



800 W BALCONY KIT

RM 16–24

off the bill a month

Yield	**~560 kWh / yr**
Per month	**≈ 47 kWh**
Kit price	**€300–500**

WORKED EXAMPLE

Aisyah · condo renter, ~490 kWh, bill ≈ RM 162

110 kWh × 36 sen → **≈ RM 40 / month · RM 475 / year**

Bill RM 162 → RM 122.40. No roof, no capex.

Sources: PVGIS SARAH3 / NASA POWER via Sunveyor, KL 1,402 kWh/kWp/yr · SoyaCincau, TNB domestic tariff structure July 2025 · RinggitPlus, Your new TNB bill explained (EEI table) · TNB press release, AFA +3.67 sen, 600 kWh exemptions (15 Sep 2026) · The Star, 800 kWh threshold to 31 Dec (Sep 2026) · Carz, Proton e.MAS 5 launch (Oct 2025) · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026) · Single Buyer, GEF projections 2025–2030 · MonsterDealz, Balkonkraftwerk 2026: costs and yield (citing Verbraucherzentrale) · Trexon, Solar carport, ~2.5 kWp per bay (2026). · ~7 km/kWh real-world EV efficiency is a team assumption (A18).

Owners earn, renters save, the public tops up.

Roof owner

e.g. Mr Lim · landed CREAM home

Roof size	≈ 4.4 kWp
Solar a month	≈ 481 kWh
× RM 0.10 / kWh	≈ RM 48
Or bill rebate	≤ 50% of bill

Result

≈ RM 58 credits

On top of the CREAM lease, plus the RM 0.90 member rate at SunD Bays.

Renter

e.g. Aisyah · condo, no roof

Slice size	1 kW
Solar a month	≈ 110 kWh
× flat credit rate	36 sen
Funded by	bay profit

Result

≈ RM 40 off

75 slices per 30-lot bay (A20). Her bill: RM 162 → RM 122.40.

Public driver

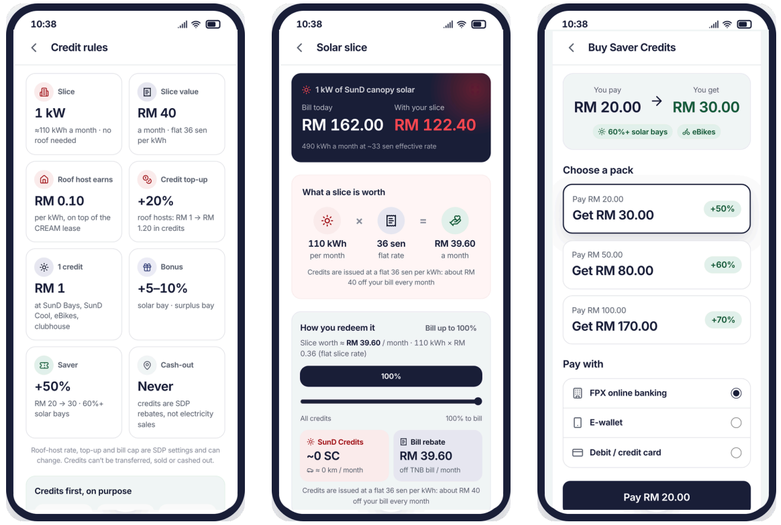
e.g. Daniel · no home charger

List price	RM 1.40 / kWh
Saver Credits	RM 20 → RM 30
Surplus bonus	+5–10%
Market DC	RM 0.88–1.80

Result

≈ RM 0.93 / kWh

Saver cash comes early but is owed as charging: a liability (A20).



Real prototype screens: credit rules, solar slice, Saver Credits.

Why RM 0.10, not 0.25? At RM 0.25 a host costs SDP ≈ RM 1,560 a year; at RM 0.10 a bay's **RM 30k host pool** covers ≈ 50–85 hosts (A20). Roof size: Ridge 1 averages 4.4 kWp a home.

Sources: SDP, Sustainability Report 2024, p.42 (May 2025) · PVGIS SARAH3 / NASA POWER via Sunveyor: KL 1,402 kWh/kWp/yr · RinggitPlus, Your new TNB bill explained (EEI table) · EV Sifu, EV charging cost Malaysia 2026

A medium bay costs RM1.2–1.7M, and chargers cost the most.

Small
10 lots

PILOT

RM 0.4–0.6M
capex before tax allowance

After GITA tax saving	**RM 0.34–0.49M**
Solar a day	**~90 kWh**
CO_2 avoided	**~24 t/yr**

Medium
30 lots

MALL

RM 1.2–1.7M
capex before tax allowance

After GITA tax saving	**RM 1.02–1.47M**
Solar a day	**~270 kWh**
CO_2 avoided	**~73 t/yr**

Large
100 lots

RM 4.0–5.8M
capex before tax allowance

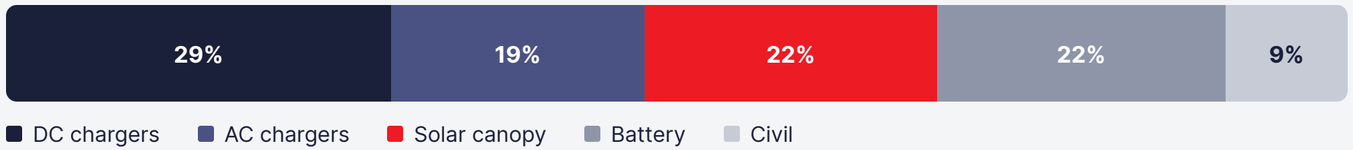
After GITA tax saving	**RM 3.4–4.9M**
Solar a day	**~900 kWh**
CO_2 avoided	**~243 t/yr**

UNIT PRICES USED

Solar canopy	**RM 3.5–5k / kWp** ASSUMPTION
AC charger 22 kW	**RM 8–15k**
DC charger 60 kW	**RM 60–80k**
Battery	**RM 1.5–2k / kWh** ASSUMPTION
Civil & metering	**+10%** ASSUMPTION

WHERE A MEDIUM BAY'S MONEY GOES

Chargers ≈ 48% of capex; panels only ≈ 22%.



Why the pilot is small

It exists to **measure the KPIs** (A17). Payback only works at **mall-level use** (A14).

NOT IN THE MODEL

- **Land** · SDP-owned
- **SunD Cool** · existing halls
- **App build** · team
- **SunD Walk** · per metre

Sources: Trexon, Solar carport, ~2.5 kWp per bay (2026) · Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Progressture Solar, ~RM2,100/kWp at 500 kWp · Trexon, EV charger installation cost 2026 (Mar 2026) · MGTC/MyHIJAU, GITA asset list (Apr 2025) · SoyaCincau, Budget 2027 cuts EV-charging GITA to 60% (Oct 2026) · PVGIS SARAH3 / NASA POWER via Sunveyor, KL 1,402 kWh/kWp/yr · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026). Battery price and civil share are team assumptions (A18).

A busy mall bay pays back in 4–9 years, before Wallet credits.

COST TO DELIVER ONE kWh · SEN



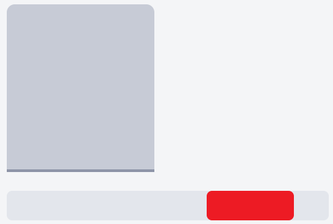
On-site solar is the cheapest kWh; blended with grid ≈ 38 sen.

LCOE = capex × 1.25 ÷ (1,402 kWh × 25 yrs). e.g. RM4,000/kWp → 14.3 sen.

MEDIUM MALL BAY · 30 LOTS · THREE SCENARIOS

1.5 sessions / lot / day

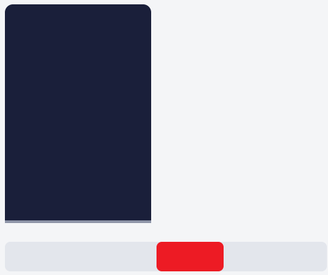
RM 165k
net cash a year, before credits



6.2–8.9 yrs payback

2.0 sessions / lot / day

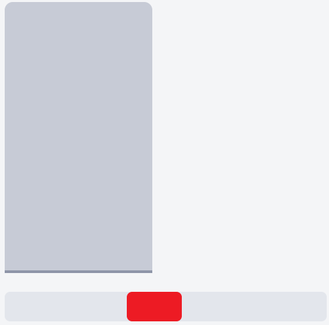
RM 216k
net cash a year, before credits



4.7–6.8 yrs payback

2.5 sessions / lot / day

RM 269k
net cash a year, before credits



3.8–5.5 yrs payback

ASSUMES 18 kWh / SESSION · 70% PUBLIC · RM1.40 / kWh · RM0.45 GRID

The 10-lot pilot does **not** pay back alone; it is a **measurement tool**.
Paybacks shown are before Wallet credits and Saver discounts: see **A20**.

Sources: Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Progressture Solar, ~RM2,100/kWp at 500 kWp · Energy Commission, SAC/CAC reduced by up to 40% (Aug 2025) · Bernama, CRESS Acceleration Package, SAC 14 sen/kWh (Sep 2026) · The Edge, CRESS SAC now 18 sen/kWh (Oct 2026) · SoyaCincau, TNB domestic tariff structure July 2025 · SoyaCincau, Gentari DC price cut (Jul 2026) · EV Sifu, EV charging cost Malaysia 2026 · PVGIS SARAH3 / NASA POWER via Sunveyor, KL 1,402 kWh/kWp/yr · MGTC/MyHIJAU, GITA asset list (Apr 2025). Session size, public share, sale and grid prices are team assumptions (A18).

SunD is priced mid-market, and lower with Saver Credits.

NETWORK (ANONYMISED)	DC · RM / kWh	AC · RM / kWh
Network K	0.88–1.20	n/a
Network D	1.00–1.30	n/a
Network C	1.13–1.50	0.90
Network G	1.30–1.80	1.05
Network T	1.25 own · 1.80 other	1.00
Network S	1.50–1.80	1.20
SunD Bay (proposed)	**1.40 list · ≈ 0.93 with Saver**	**residents: credits**

Values: EV Sifu 2026 table. Public rates vary by site and change often.

Market signals

- 30,000 public chargers targeted by 2030
- 6,904 public bays today (Jul 2026)
- National charger use is still low: busy bays win

Why SunD fills bays

Credit holders already live next door, so a SunD Bay starts with users on day one: the weak point of today's networks.

KEY POINT SunD lists at a mid-market RM 1.40/kWh, and **≈ RM 0.93 with Saver Credits**: near the cheapest DC price in the market.

Sources: EV Sifu, EV charging cost Malaysia 2026 · Lowyat.NET, 6,416 public chargers, 30,000 target (Jul 2026) · Business Today, 6,904 public charging bays (Sep 2026)

Elmina first, then every SDP township and new launch.

WEEKS 1–8

Build

Wallet live in one Elmina precinct · 10-lot pilot bay

MONTHS 3–6

Pilot

7 KPIs from week 1 · go / no-go at month 6

MONTHS 6–12

Extend

30-lot mall bay, open to all · Saver Credits · first SunD Walk

YEAR 2+

Scale

Existing SDP townships · every new launch · eBike · licensing · SunD sub-brand

Not limited to the township

One wallet · every SunD location · every driver

Elmina pilot
1 precinct · 10-lot bay

›

SD townships
existing + every new launch

›

Malls · offices · campuses
any site, own canopy

›

Highways · petrol stations
charge on the go

›

Other developers
license SunD

Pilot success = RM saved per subscriber · 25–35% bay solar share · ≥ 40% bay utilisation (KPIs 1–3, A17)

Sources: SDP, City of Elmina · SDP, Integrated Annual Report 2025 (Apr 2026)

Seven KPIs give SDP a measured yes or no at month 6.

01 PEOPLE

RM saved per subscriber a month

NOW RM 0 → MONTH 6 RM 40

Wallet ledger + smart-meter tariff

02 PLANET

Bay kWh from on-site solar

NOW ≈ 0% → MONTH 6 25–35%

Charger + inverter meters

03 COMMERCIAL

Bay utilisation

NOW n/a → MONTH 6 ≥ 40%

Charger session logs

04 PEOPLE

SunD Cool visits per heat-alert day

NOW 0 → MONTH 6 Tracked

Wallet check-ins on alert days

05 PEOPLE

How much cooler SunD Walk feels

NOW 4-wk log → MONTH 6 ≥ 3 °C

Sensors under vs beside canopy

06 PLANET

Tonnes CO_2 avoided

NOW 0 → MONTH 6 ~2 t / month

Solar kWh × 0.74 kg

07 PEOPLE

Wallets that are roofless

NOW 0 → MONTH 6 ≥ 50%

Wallet registrations

DECISION AT MONTH 6

Go / no-go on KPIs 1 · 2 · 3

Needs from SDP / TNB: CREAM host output, tariff bands (with consent), charger logs, hall entries.

Sources: RinggitPlus, Your new TNB bill explained (EEI table) · Reuters/GSO via The Edge, 2025 generation mix (Jan 2026) · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026) · DOSM, Current Population Estimates 2026 (Jul 2026) · Xu, Bardhan et al., Sci. Total Environ. 958:177864 (2025) · Bernama, 17 areas under Level 1 heat status (Aug 2026) · SDP, Sustainability Report 2024 (May 2025).

Our 13 assumptions, and what moves if each is wrong.

8 numbers move payback

Watch these first.

COST capex and running cost

Canopy capex — RM3.5–5k / kWp
If wrong: ±RM500 = ±RM38k capex

Battery capex — RM1.5–2k / kWh
If wrong: ±RM500 = ±RM90k capex

Civil share — +10%
If wrong: at 20%, payback +0.5–0.8 yr

O&M — RM100 / day
If wrong: doubled, −RM36k a year

REVENUE price and usage

Session size — 18 kWh
If wrong: ±6 kWh = ±⅓ revenue

Public share — 70%
If wrong: at 50%, payback ≈ 10–14 yrs

Sale price — RM1.40 / kWh
If wrong: ±10 sen = ±RM28–34k a year

Grid top-up — RM0.45 / kWh
If wrong: ±5 sen = ∓RM10–20k a year

5 numbers do not touch payback

They change savings, CO_2 or a single example.

OTHER EFFECTS savings, CO_2, examples

Solar yield SAVINGS — 3.6 kWh/kWp/day
If wrong: at 3.84, slice worth +7%

Grid emission factor CO_2 — 0.74 kg / kWh
If wrong: the 2030 value cuts tonnes by 36%

Host roof output TO MEASURE — ~16 kWh / day
Next: metered for pilot hosts (4.4 kWp)

Balcony kit yield TO MEASURE — 560 kWh / yr
Next: measured in the sandbox

EV efficiency ILLUSTRATION — 7 km / kWh
If wrong: only the “60 km” example changes

KEY POINT These 13 are the model inputs; sessions/lot/day is in A14 and the credit levers in **A20**. 8 move payback, so the pilot’s **price and usage data** matter most.

Bases: PVGIS SARAH3 / NASA POWER via Sunveyor, KL 1,402 kWh/kWp/yr · Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Trexon, EV charger installation cost 2026 (Mar 2026) · Carz, Proton e.MAS 5 launch (Oct 2025) · SoyaCincau, Gentari DC price cut (Jul 2026) · EV Sifu, EV charging cost Malaysia 2026 · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026) · Single Buyer, GEF projections 2025–2030 · SDP, Sustainability Report 2024 (May 2025) · MonsterDealz, Balkonkraftwerk 2026: costs and yield (citing Verbraucherzentrale)

Ten questions judges may ask, answered.

Is it allowed?

rules, permits and partners

Q1
Can you legally “pay” people in electricity?

✓ We don’t sell power
Credits are **SDP-funded rebates**; TNB stays the supplier.

Q2
Isn’t the 5 km CREAM radius a problem?

✓ No CREAM transfer
Bay power is **on-site solar**; the grid tops up the rest.

Q3
What if Solar ATAP rejects a canopy?

✓ Zero-export route
Malls are multi-tenant, so **zero-export** mall bays (A21).

Q4
Balcony solar isn’t legal.

✓ Correct, so 2027
It is a **2027 sandbox proposal**, not in the pilot. Households buy kits; renters need landlord OK.

Q5
What happens to CREAM roof hosts?

✓ Nobody worse off
Lease unchanged; SunD credits are **extra**.

Does it work and pay?

money, timing and demand

Q6
Who pays for the credits?

✓ Bay profit
Charging profit funds **75 slices + host pool** (A20); Saver cash is a liability.

Q7
Does a bay pay back?

✓ Mall bay: 4–9 yrs
The pilot is a measurement tool; a **mall bay pays back** (A14), before credits (A20).

Q8
Solar peaks at midday; people charge at night.

✓ Battery + nudges
Battery stores midday solar; nudges shift charging to **11am–3pm**.

Q9
Will anyone use a cool room?

✓ We measure it
Visits are a KPI from zero; **+12.9%** elderly admissions per +1 °C.

Q10
Why not just sell GET / RECs?

✓ Misses residents
GET and RECs never put solar **in residents’ hands**.

Evidence: SDP, Residential solar launch under NETR (Sep 2025) · EV Sifu, EV charging cost Malaysia 2026 · The Edge, Govt sets guidelines for rooftop solar leasing (Mar 2025) · Energy Commission, Guidelines for Solar ATAP (in force 1 Jan 2026) · Yong et al., IJERPH 20(10):5910, heatwaves and Selangor elderly hospitalisations (2023) · The Edge, GET premium cut to 3–5 sen/kWh (Jul 2025) · S&P Global, Malaysian solar I-REC US$3.45/MWh (Apr 2026) · Bundesnetzagentur MaStR via kleineskraftwerk.de, 1.53 M systems (Oct 2026) · CAP via Aliran, Balcony solar panels for clean energy transition (27 Aug 2026) · The Star/Bernama, EC eyes 2027 regulatory sandbox (Jun 2026) · SDP, Community solar rollout with TNB (Apr 2025).

Bay charging profit pays for slices and host rewards.

ONE 30-LOT MALL BAY · A YEAR · 2.0 SESSIONS/LOT/DAY

Public charging RM 1,058 a day	**+RM 386k**
Grid top-up and upkeep RM 465 a day	**−RM 170k**
Bay cash before credits RM 593 a day	**RM 216k**
75 solar slices 75 × RM 40 × 12	**−RM 36k**
Roof-host pool ≈ 50–85 hosts	**−RM 30k**

A slice is 1 kW of SunD canopy solar for one family: 110 kWh a month credited at a flat 36 sen. Hosts are the CREAM homes nearest the bay.

Bay cash after Wallet credits

RM 150k a year

Payback **6.8–9.8 yrs** after GITA (capex RM 1.02–1.47M), vs 4.7–6.8 yrs before credits (A14).

USAGE MOVES IT MOST

SESSIONS	BEFORE CREDITS	AFTER CREDITS	PAYBACK AFTER
1.5	RM 165k	RM 99k	10.3–14.8 yrs
2.0 base	RM 216k	RM 150k	**6.8–9.8 yrs**
2.5 fleet	RM 269k	RM 203k	**5.0–7.2 yrs**

SAVER DISCOUNT SENSITIVITY

Saver: pay RM 20, get RM 30 ⇒ **RM 0.93/kWh** instead of RM 1.40.

No public kWh on Saver	**RM 593 a day**
30% of public kWh on Saver	**≈ RM 487 a day**
All public kWh on Saver	**≈ RM 240 a day**

Prepaid Saver cash is owed as charging (a liability), so it is not counted as funding.

WHAT A CREDIT REALLY COSTS SDP

A credit spent at a SunD service costs SDP only the kWh behind it. A host's **RM 58** of credits buys 64 kWh at the RM 0.90 member rate; that is 64 × RM 0.45 grid ≈ **RM 29**.

A bill rebate is cash: **RM 48 costs RM 48**. A slice is a flat 36 sen credit per kWh, 110 kWh ≈ **RM 40**.

LEVERS SDP SETS

ASSUMPTION

- Slice rate **36 sen/kWh**
- Slices per bay **75**
- Host rate **RM 0.10/kWh**
- Credit top-up **+20%**
- Host pool **RM 30k/yr**
- Member rate **RM 0.90/kWh**
- Saver bonus **+50%**
- Tax rate (GITA) **24%**
- Panel life **25 yrs**
- Upkeep **RM 100/day**

KEY POINT Credits are paid from bay profit, so they grow bay by bay: **Elmina first**, then other SDP townships and every new launch.

Bases: Team model, A11–A14 (ASSUMPTION levers listed above) · RinggitPlus, Your new TNB bill explained (EEI table) · EV Sifu, EV charging cost Malaysia 2026 · SoyaCincau, Budget 2027 cuts EV-charging GITA to 60% (Oct 2026)

Malls need zero-export; SDP-owned sites can use Solar ATAP.

Solar ATAP

IN FORCE · 1 JAN 2026

Self-consumption with export credit

Who	one consumer per premise; **multi-tenant excluded** (cl. 6.1(d), cl. 2)
Where	rooftops; car parks and walkways in the same compound (cl. 7.1–7.2)
Size	non-domestic ≤ 100% of max demand, ≤ 1,000 kW (cl. 8.3, 8.5)
Export	offsets imports; non-domestic at average SMP; no carry-forward (cl. 3.6.1)

Fits: SDP clubhouse, community hall, sales gallery or office car park.

Zero-export self-consumption

SELCO · 2026

Use every kWh on site, export nothing

Who	the owner or occupier; **a tenant can be the consumer** (cl. 2, 3.1(a))
Where	within the same premises (cl. 3.1(d)); canopy to confirm with EC
Size	no cap for non-domestic (cl. 3.4(a)); battery rule only above 1 MWac (cl. 3.6)
Export	none at any time; anti-export device (cl. 3.4, 5.5)

Fits: a mall SunD Bay on its own TNB meter.

WHY ZERO EXPORT COSTS A BAY ALMOST NOTHING

4× the bay uses **1,080 kWh** a day; its canopy makes **270 kWh**.

Every solar kWh is used on site and the 180 kWh battery takes the midday peak, so export credit adds little.

SITE BY SITE

Mall bay	**zero-export, own meter**
Clubhouse · community hall	**Solar ATAP cl. 7.2**
SunD Walk	**on an SDP premise's meter**
Condo car park	**JMB/MC decision**

Confirm in the pilot with TNB/EC/SEDA: bay meter account, canopy as a same-premises structure, anti-export device, walkway host premise.

KEY POINT A mall is multi-tenant, so its bay runs **zero-export on its own meter**; it uses four times what its canopy makes anyway.

Sources: Energy Commission/SEDA, Guidelines for Solar ATAP (in force 1 Jan 2026) · Energy Commission, Guidelines for Solar PV for Self-Consumption, Peninsular Malaysia (2026) · Team model, A5 and A14

More cash per bay, beyond the base case.

Fleet and e-hailing night deals

IN OUR TABLE

+RM 53k a year

Fill lots after dark: **2.0 → 2.5 sessions** per lot per day. Payback after credits falls from 6.8–9.8 to **5.0–7.2 yrs** (A20).

Time-of-day pricing

NOT MODELLED

Charge more in the evening peak, when the battery is empty and the kWh comes from the grid; keep the **midday solar discount**.

Idle fee after a full charge

NOT MODELLED

A per-minute fee once charging ends frees the lot for the next driver, so **more sessions** per lot.

SunD Pass membership

NOT MODELLED

Public drivers pay a monthly fee for a lower per-kWh price: **recurring cash** that can replace part of the Saver discount.

Merchant offers in the Wallet

NOT MODELLED

Drivers wait while they charge at the mall; **merchants pay** to reach them with offers in the app.

SunD Balcony kits

FROM 2027

After the EC sandbox, SDP **sells and installs** certified 800 W kits; households buy them (€300–500 a kit in Germany).

SunD-ready new launches

NOT MODELLED

93% of Malaysians are willing or open to paying more for homes with green features; no premium % is published.

WHERE THEY APPLY

Elmina first, then other SDP townships and every new launch.

Each lever is tested in the pilot before it enters the base case.

KEY POINT Only the **fleet lever** is in our numbers; the other six are upside the pilot can test.

Sources: Team model, A14 and A20 · Business Today, PropertyGuru 1H26 Consumer Sentiment Study (Sep 2026) · MonsterDealz, Balkonkraftwerk 2026: costs and yield (citing Verbraucherzentrale) · The Star/Bernama, EC eyes 2027 regulatory sandbox (Jun 2026)

Where the main deck simplifies, the precise version.

SLIDE	MAIN DECK	PRECISE VERSION	PAGE
S1 · S4 · S8	**“RM 40 off a month”**	A slice credits 110 kWh at a flat 36 sen ≈ RM 40, paid from bay profit: 75 slices per bay.	A11 · A20
S2	**“22.8%” and “37%” high-rise**	Derived from DOSM Census 2020 living-quarters counts.	A1
S2	**“+1 °C → +12.9% admissions”**	Mean apparent temperature, people aged 60+, Selangor (Yong et al. 2023).	A1
S2	**“~1M homes past 600 kWh”**	Reported by Bernama (Sep 2026).	A1
S3	**“18 : 1”**	Battery EVs per public charging bay: 122,286 ÷ 6,904.	A1
S3 · S8	**“Balcony kits from 2027”**	A proposed EC sandbox; households buy kits; renters need landlord or JMB consent.	A9
S4	**“Wallet shares the solar”**	Residents receive credits; the power and its green attributes stay with the CREAM off-taker.	A8 · A12
S4 · S5	**“Solar first, grid tops up”**	At a busy mall bay the canopy covers ≈25% of the energy; the grid supplies ≈75%.	A5 · A14
S4 · S8	**“−3.1 °C under shade”**	Street shade cut feels-like (UTCI) by 3.1 °C in a Singapore study; SunD Walk target ≥ 3 °C (KPI 05).	A17
S5	**“CRESS 23–27 sen”**	Firm supply with a battery; solar-only CRESS ≈ 47–49 sen (40 sen non-firm access).	A10 · A14

SLIDE	MAIN DECK	PRECISE VERSION	PAGE
S5	**“First in Malaysia”**	We found no Malaysian scheme that links residents, credits and charging this way.	—
S8	**“−40% Scope 1 & 2”**	On-site solar avoids ≈ 73 t CO_2 a year per bay; the net effect of grid top-up is still to confirm.	A13
S8–S10	**“RM 593 a day · 4.7–6.8 yrs”**	Before Wallet credits and Saver discounts; after credits 6.8–9.8 yrs.	A14 · A20
S9	**“Cheapest kWh in town”**	Canopy kWh costs 12–18 sen; blended with grid ≈ 38 sen at the base case.	A14
S9	**“93% would pay more”**	Willing or open to paying more for homes with green features; no premium % published.	A2
S9 · S10	**“Elmina pilot”**	Elmina is the first site; the model applies to existing SDP townships and every new launch.	A16
S10	**“GITA 100% → 60%”**	GITA is a tax allowance; 60% for chargers from 2027 is a Budget 2027 proposal.	A8 · A13
S10	**“Credits: no new licence”**	True for SDP-funded rebates; paid Saver packs need a BNM e-money check.	A8
S10	**“Solar ATAP allows car parks”**	Only on single-consumer premises; mall bays run zero-export on their own meter.	A21
S10	**“Demand risk: managed” · “3–4 mo”**	Payback depends on sessions per lot per day; build time is a team estimate.	A6 · A14

KEY POINT Every simplification in the main deck maps to a **precise, sourced line** in this appendix.

Sources: Xu, Bardhan et al., Sci. Total Environ. 958 (2025) · Business Today, non-firm SAC remains 40 sen (Sep 2026) · SoyaCincau, Budget 2027 cuts EV-charging GITA to 60% (Oct 2026) · SDP, CREAM pilot in City of Elmina (Sep 2025) · DOSM, Special Report: Living Quarters, Census 2020 (Apr 2026) · Yong et al., IJERPH 20(10):5910